

Software Requirements Specification (SRS)

Steel Microstructure Classification Using Convolutional Neural Networks (CNN)

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1. Introduction

This project develops a CNN model to classify steel microstructure images into alloy categories.

2. Problem Statement

Build an image classification system for steel alloy microstructures.

3. Objectives

Dataset preparation, preprocessing, CNN training, evaluation, and Streamlit deployment.

4. Functional Requirements

Upload image, preprocess, predict class, and display result.

5. Non-Functional Requirements

Performance, reliability, usability, scalability, and maintainability.

6. Tools and Technologies

Python, TensorFlow/Keras, NumPy, Pandas, Matplotlib, Kaggle, Streamlit, GitHub.

7. Dataset Description

CPJ Alloys, HR Alloys, and P92 Alloys steel microstructure image classes.

8. System Design Overview

Input Image → Preprocessing → CNN → Prediction → Result

9. Model Evaluation

Accuracy, confusion matrix, precision, recall, and F1-score.

10. Conclusion

CNN-based steel microstructure classification system.